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ORIGINAL INVESTIGATION

Flight Path and Scheduling Effects on Perceived Jet Lag, Fatigue, and Sleep in Footballers Traveling to and From National Teams

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Purpose: This study examined posttravel perceptual responses of national-team footballers (soccer) following different flight paths, arrival/departure times, and trip contexts. ***Methods:*** Details of 396 flights from national-team players (N = 68) were obtained and verified via an online flight database. Each player provided ratings of perceptual fatigue, sleep, soreness, stress, and jet lag for 2 days before and after each trip. The flight path (continents of departure and arrival), travel context (into vs out of national team), and arrival and departure times were obtained for each trip. Linear mixed models compared the pretravel with posttravel change in perceptual responses based on flight path, context, and schedule. ***Results:*** Perceived jet-lag ratings were more responsive to travel variables ($R^2 = .48$) than other perceptual ratings ($R^2 < .26$). Travel from Asia to Europe ($P < .05$) and Europe to Australia ($P < .001$) had significantly higher jet-lag ratings than all other paths. Fatigue scores were worst following Asia to Europe ($P < .05$) and Europe to Australia ($P < .05$) travel, while sleep scores were worst following Europe to Australia travel ($P < .01$). Perceptual responses were poorer following travel from national team to club compared with all other travel contexts ($P < .05$). Arrival during the daytime (11 AM to 5 PM) resulted in better perceptual responses than early-morning or late-night arrivals ($P < .05$). ***Conclusions:*** Perceived jet-lag ratings are more responsive to travel demands than perceptual wellness scales in national-team football athletes. Poorer perceptual responses may be expected when travel is longer in nature, ends later in the day, or involves travel out of the national team back to club.

Keywords: travel, transition, soccer

National football (soccer) teams often require athletes to undertake extensive travel from different club locations into a single competition or camp location. The diversity of travel for athletes spread around the world results in a range of travel-induced states on arrival at camp.[1] Factors that influence this posttravel state, such as jet lag, travel fatigue, or sleep disruption, are dependent on the travel duration, time-zone change, and direction, which vary based on individual travel schedules.[1,2] Hence, practitioners need to consider the effects of these schedules on athlete arrival into the national team and return to club. Further, the flight path, arrival/departure time, and whether the trip was into or from the national team, are all likely to influence the athlete's response to the journey.[1] While travel research uses extensive jet lag questionnaires, physiological, or performance measures,[3] these are logistically impossible in national team contexts, where player monitoring is commonly limited to perceptual questionnaires relating to fatigue, sleep, soreness, and stress.[4] Despite concerns about the validity of these "wellness" scales,[5] they capture elements of symptoms reported in jet lag and travel fatigue,[1] though few studies assess their responsiveness to different travel demands. Given the pervasive use of these scales in football teams, understanding their responses to travel across different paths, schedules, and contexts can inform athlete monitoring of travel for national football teams.

For many non-European national teams, travel often follows particular patterns, whereby players located in a range of countries will travel common paths based on club and competition locations.[6] Prior research has identified a detrimental effect of time zone change and direction of travel on jet lag/travel fatigue symptoms following arrival in the club or national team.[2,7,8] Specific flight details for national teams are often determined at the last minute, and therefore only the general location an athlete is traveling to and from is known to practitioners until the trip is finalized. However, as many flight paths are commonly traveled for National Team duties, this information can still provide useful insight. While current travel studies have explored fatigue, jet lag, and sleep responses in footballers following trips between Australia, Asia, Europe, South America, and North America,[9–16] such studies only explore singular trips and deeper understanding of responses to common flight paths is missing. Given the variation in populations and methodologies between studies, drawing comparisons on the effects of different flight paths on fatigue, sleep, and jet lag responses is difficult.[8,14,16,17] As players involved in any national team camp are required to travel from various locations, understanding travel responses to common flight paths can aid travel management strategies. Furthermore, effects of the path may also be influenced by whether it involved traveling into or out of the national team.[4] While poorer countermovement jump, jet lag, and fatigue were reported after both outbound and return travel responses from a 6-hour flight, no comparisons were made between the 2 trip contexts and only a single short-duration trip was reported.[17] Larger data sets on longer travel are missing to inform national team footballers undertaking both outbound and return travel to clubs.

Further concerns for national team players include the departure/arrival time and how these factors may influence sleep and fatigue in the days following arrival. Arrival closer to sleep periods has previously been related to reduced jet lag and fatigue ratings in elite athletes.[18] However, these results consider only a single trip and do not cover the diversity of travel schedules experienced by national team footballers. Broader comparisons are needed across

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the range of arrival and departure times experienced by national team footballers. Accordingly, for a national football federation, understanding how trip factors influence player responses to travel will allow staff to better plan for athlete arrival/departure. This study aims to compare posttravel perceptual jet lag, fatigue, sleep, soreness, and stress ratings between different flight paths, schedules, and trip contexts.

Methods

Participants

Participants included 68 professional footballers (soccer) from a senior men's national team who were part of traveling squads between March 2018 and July 2022. Consent to use the data anonymously was obtained from the national football federation. All athletes provided consent for the collection and use of their data anonymously via national team contracts. Ethical approval was obtained from the institutional Human Ethics Committee (ETH20-5080).

Overview

The details of 796 flights were obtained and aligned with pretravel and posttravel perceptual responses. Overall, 396 flights included pretravel and posttravel perceptual scales, and 223 flights included perceived jet lag ratings. Perceptual ratings of fatigue, sleep, soreness, and stress (collectedly termed "wellness") were obtained for 2 days prior to travel and up to 3 days after travel. Perceived jet lag ratings were obtained on the first 3 days after arrival from travel. All measures were obtained as part of national team monitoring procedures with players required to complete a daily perceptual questionnaire via the organization's athlete monitoring software on the athlete's smartphone. All players had previously used the questionnaire extensively. Trips were excluded if they did not include at least one pretravel and all posttravel perceptual monitoring responses.

Travel Details

Travel details for each trip were obtained from booked travel schedules, with the arrival and departure times for each trip then verified using an online flight database (Flightera.com). Trips were classified based on (1) flight path, (2) context, and (3) arrival and departure time. The flight path was classified based on the departure and arrival continent, based on the geographical location of the airport (not including land-based travel). Accordingly, the following categories were derived: (1) Asia to Asia, (2) Asia to Australia, (3) Asia to Europe, (4) Australia to Asia, (5) Europe to Asia, and (6) Europe to Australia. The context of the trip was classified based on whether travel occurred into a national team camp (outbound), between 2 different locations within the same camp (transition), or out of the national team (return). The arrival and departure time of each trip was grouped into categories of morning (05:00–11:00), daytime (11:00–17:00), evening (17:00–23:00), and night (23:00–05:00). A player's age and number of national-team appearances at the time of travel were also obtained from the federation databases and included in the analysis.

Perceptual Response Scales

Players completed a perceptual questionnaire every morning from 2 days before travel into national team through to 3 days after they left the national team. In this questionnaire, players provided subjective ratings of fatigue, sleep, soreness, and stress via a 7-point Likert scale. Descriptive anchors were included at scores of 1, 4, and 7, with scores of 1 labeled as having "no" fatigue, soreness, or stress and "outstanding" sleep. Scores of 4 labeled as "moderate" fatigue, soreness, or stress and "average" sleep. Scores of 7 labeled with "maximal" fatigue, "extreme" soreness, "worst possible" stress, or "horrible" sleep. The sum of all 4 scales for each day was included in analysis as a "total wellness" score. For each trip, raw scores were converted into a change score by subtracting the latest score obtained prior to departure from the score on each day (day 1 and day 2 postarrival).

These perceptual monitoring scales are frequently used in football teams to monitor responses to training, especially given the lack of available objective data for many national teams.[4] Although these scales have been suggested to lack a conceptual framework,[5] prior studies observed their responsiveness to training stress in both national[19] and club football teams.[20] However, the limitations of these scales should be considered when interpreting results as travel may account for only a small proportion of variation in scores—which further necessitates the current study. Despite this, these scales represent a practical and frequently used tool in national teams to monitor athletes and can potentially aid understanding of travel responses in national teams.[4]

Perceived Jet-Lag Rating

Athletes completed a perceived jet lag rating every day for 3 days after travel. A modified version of the single-item jet lag rating from the Liverpool John Moore's University Jet Lag questionnaire[21] was used. Athletes were asked "Do you have any jet lag or fatigue from your travel?" and answered on a 10-point rating scale with scores of 0 labeled as "none at all" and scores of 10 labeled as "extreme." While jet lag is a bio-psychological and chronobiological concept, the Liverpool John Moore's University Jet Lag questionnaire attempts to measures this perceptually, and the decision to include "travel fatigue" in this study was due to the inability to distinguish between symptoms of the 2 conditions. As such, this scale aimed to be a more specific measure of travel response compared with the aforementioned perceptual scales. For each trip, perceived jet lag ratings were obtained as a raw value and labeled by the day they were collected relative to arrival (ie, +1, +2, +3).

Statistical Analysis

Travel details and perceptual scales were collated into a single excel spreadsheet and imported into R studio.[22] Perceptual monitoring scores for each day were labeled as day 1 (D1), day 2 (D2), or day 3 (D3-Perceived jet lag only) based on when the score was provided relative to arrival. Each outcome was aligned with the details of the prior travel. For all statistical tests, statistical significance was set at .05.

To analyze the influence of travel factors on the perceptual response to travel, linear mixed models were built for each outcome using the lme4 package.[23] A numerical player identifier was included as a random effect within the model to account for nonindependence of outcomes. Models were built using a stepwise approach with the inclusion of fixed effects determined by statistical significance as measured by an F test with Satterthwaite degrees of freedom approximation.[24] The models Akaike Information Criterion and R^2 values were used to determine the overall fit of

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the model at each step. Once the final model had been built, assumptions of normality and homogeneity of variance were checked using Q–Q plots and residual plots. Post hoc pairwise comparisons between categorical variables were performed via estimated marginal means.[25] Given the absence of significant interactions between variables, the mean value for each category was averaged out over levels of other variables.

Results

Model Details

Details of the final models for each outcome variable are provided in Table 1. Based on R^2 values, perceived jet lag scores were most sensitive to the flight path, context, and arrival time ($R^2 = .48$). Conversely, perceptual "wellness" scales showed lower sensitivity ($R^2 = .15$–$.26$) with fatigue showing the highest association with analyzed travel variables.

Perceived Jet Lag

Flight paths from Asia to Europe ($P = .002$–$.047$) and Europe to Australia ($P < .001$) produced significantly higher jet lag ratings than all other paths (Figure 1A). Travel from Australia to Asia resulted in lower perceived jet lag ratings than travel within Asia ($P < .001$). Figure 1B shows perceived jet lag was significantly higher following transition travel compared with travel into the national team ($P < .001$); however, no other differences were observed between trip types. Player age had a positive relationship with jet lag scores, with an increase in 1 unit (year) resulting in a 0.185 increase in perceived jet lag score ($P < .001$). In contrast, for each national team appearance, a player's perceived jet lag score decreased by 0.023 ($P = .008$).

Effects of Flight Path on Perceptual Wellness Scales

Flight path had a significant effect on all perceptual scales (total wellness $P = .005$, fatigue $P < .001$, sleep $P < .001$, soreness $P = .013$, and stress $P = .036$). Pairwise comparisons between each flight path are shown in Figure 2. Europe to Australia travel resulted in higher (poorer) wellness compared with all other flight paths except Asia to Europe ($P < .01$). Similarly, for fatigue ratings, Europe to Australia had significantly higher scores than Asia to Asia, Australia to Asia, and Europe to Asia ($P = .004$–$.043$). Asia to Europe travel resulted in higher fatigue scores compared with Asia to Asia and Australia and Europe ($P = .003$–$.047$). Travel from Australia to Asia also caused higher fatigue scores than travel from Asia to Asia ($P = .013$). Higher perceptual sleep ratings were observed after Europe to Australia compared with all other flight paths ($P < .01$). Significantly higher changes in soreness scores were observed after both Europe to Australia and Asia to Europe travel compared with Asia to Asia ($P = .068$; $P = .045$) and Asia to Australia ($P = .041$; $P = .006$). Finally, lower stress ratings occurred after travel from Europe to Asia compared with Asia to Australia ($P = .014$), Asia to Europe ($P = .004$), and Australia to Asia ($P = .020$).

Effects of Trip Type on Perceptual Wellness Scales

Figure 3 shows higher scores were observed following return travel compared with outbound travel and transition travel for total wellness (into $P < .001$; transition $P < .001$), fatigue (into $P = .002$; transition $P < .001$), sleep (into $P = .024$; transition $P < .001$), and soreness (into $P = .003$; transition $P = .004$). For stress ratings, higher scores were observed after transition travel compared with travel into national team ($P = .008$).

Effects of Arrival Time on Perceptual Wellness Scales

Compared with morning arrivals, daytime arrivals were associated with significantly lower total wellness ($P < .001$), fatigue ($P = .006$), sleep ($P = .050$), and soreness scores ($P < .001$). Daytime arrivals also resulted in lower total wellness ($P = .008$) and soreness ($P = .020$) compared with night arrivals. Similarly, evening arrivals resulted in lower total wellness ($P = .029$), fatigue ($P = .013$), and soreness ($P < .001$) scores compared with morning arrivals. However, compared with daytime arrivals, evening arrivals had higher total wellness ($P = .005$) and sleep scores ($P = .002$). Lastly, stress scores were the highest after night arrivals compared with all other arrival times (morning $P = .008$, daytime $P = .003$, and evening $P = .005$; Figure 4).

Discussion

The current study identified travel-induced perceptual responses of jet lag, fatigue, sleep, soreness, and stress from elite national team footballers based on flight paths, arrival/departure times, and trips contexts. Travel from Europe to Australia or Asia to Europe had the greatest impact on athlete perceptual responses. Travel responses were worst when leaving the national team compared with travel into the national team or transitioning between national team locations. Arrival during the day (between 11:00 and 17:00) resulted in better perceptual responses. While athlete ratings of fatigue, sleep, soreness, and stress are responsive to certain travel

Table 1 Final Models Detailing the Relationship Between Travel Scheduling Factors and Perceptual Measures

Model	Akaike information criterion	R^2	Fixed R^2
Total wellness ~ Flight path + Day + Trip type + Arrival time + (1|Player code)	3445.41	.23	.17
Fatigue ~ Flight path + Day + Trip type + Arrival time + (1|Player code)	2038.65	.26	.21
Sleep ~ Flight path + Day + Arrival time + Trip type + (1|Player code)	2426.79	.22	.13
Soreness ~ Flight path + Arrival time + Trip type + Day + (1|Player code)	1963.19	.16	.13
Stress ~ Flight path + Departure time + Arrival time + Trip type + (1|Player code)	1193.08	.15	.07
Perceived jet lag ~ Day + Flight path + Trip type + Player age + National team caps + (1|Player code)	2380.37	.48	.31

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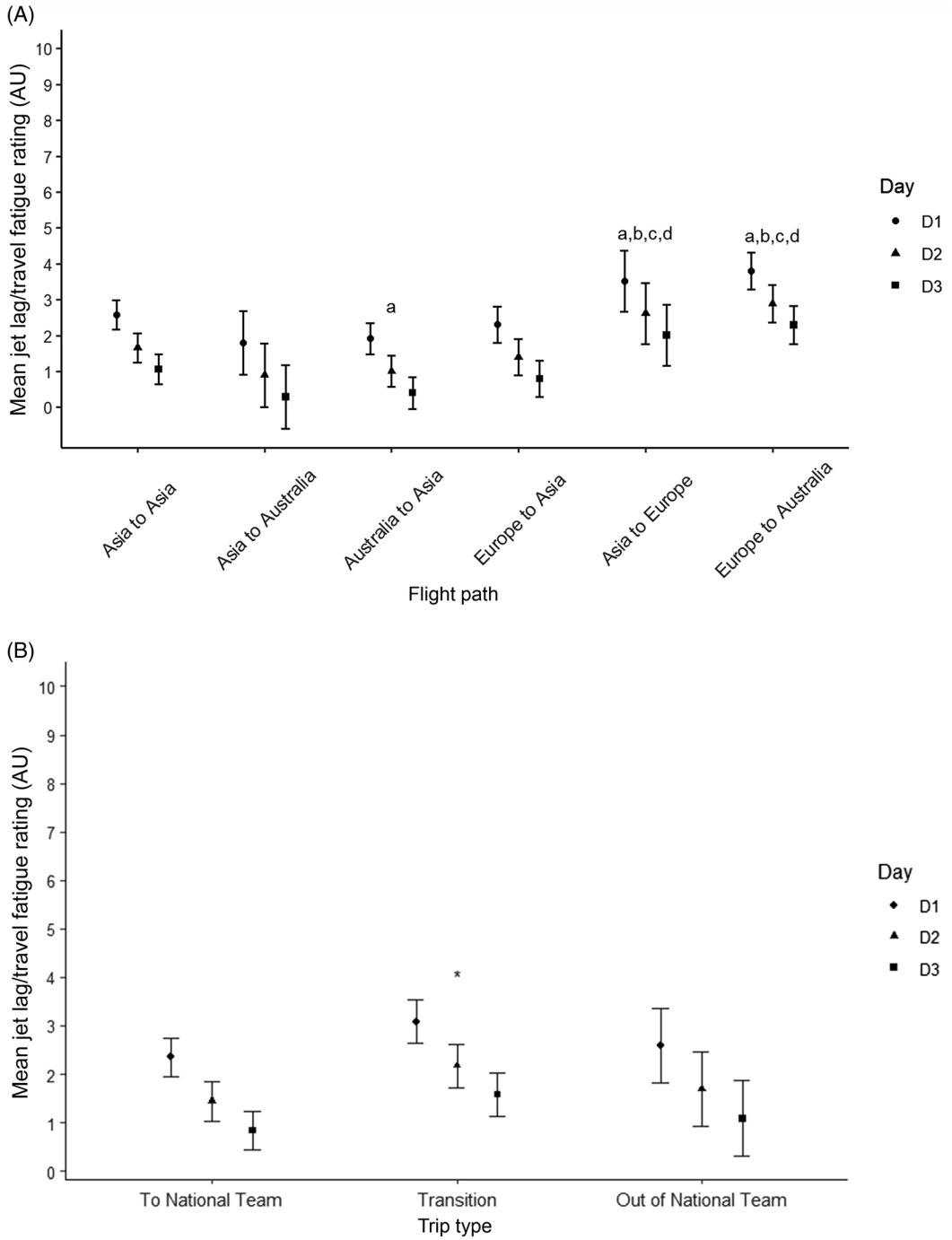

Figure 1 — Mean perceived jet-lag score by flight path and trip type (averaged out over day, player age, and national-team camps). [a]Significantly different from Asia to Asia. [b]Significantly different from Asia to Australia. [c]Significantly different from Australia to Asia. [d]Significantly different from Europe to Asia. *Significantly different from travel to national team.

demands, a subjective jet lag scale represents a more responsive tool to monitor travel responses.

Importantly, this study showed that a perceptual jet lag rating has better association with variations in travel demands than perceptual wellness measures. Although the full LJMUJLQ scale[18] represents a more validated tool to monitor travel responses, this study highlighted a simplified version can be a practical and informative tool for national football teams. As expected, jet lag was highest following trips from Europe to Australia (Figure 1), representing eastward travel with the largest time zone change.[1] Prior studies show support, with detrimental jet lag symptoms following longer travel demands and eastward travel.[2,7,8] Interestingly, elevated jet lag was also observed following Asia to Europe travel. As this flight path was common for athletes returning to their clubs, it is possible that accumulated travel fatigue from multiple long-haul flights in a short time may explain this,[26] though further research is needed. A limitation of these comparisons, however, was that due to match scheduling and COVID-related venue changes, insufficient data existed from Australia to Europe trips, and thus such trips were not included. Regardless, support in the

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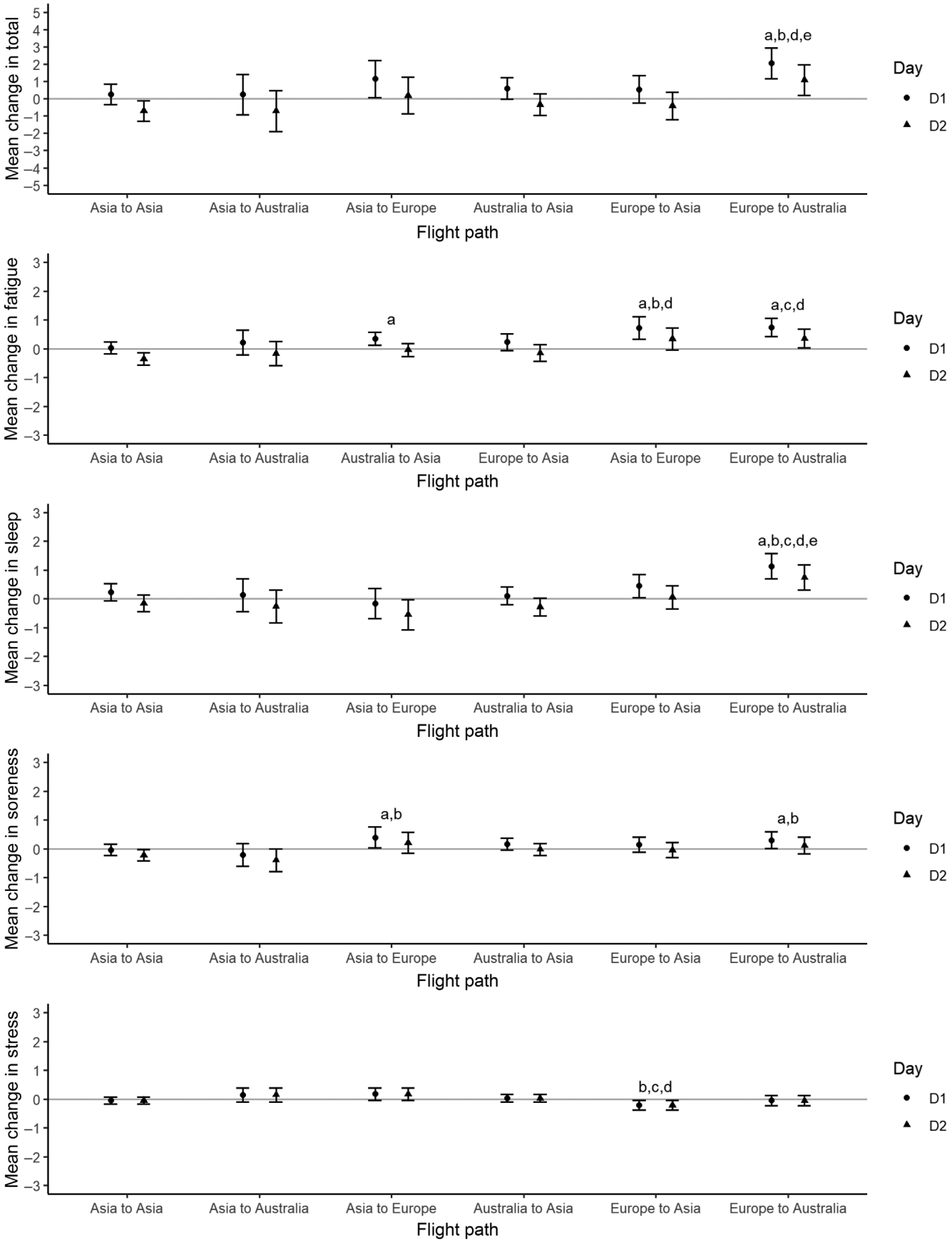

Figure 2 — Mean change in perceptual wellness scores by flight path (values are averaged out over levels of other variables). [a]Significantly different from Asia to Asia. [b]Significantly different from Asia to Australia. [c]Significantly different from Australia to Asia. [d]Significantly different from Europe to Asia. [e]Significantly different from Asia to Europe.

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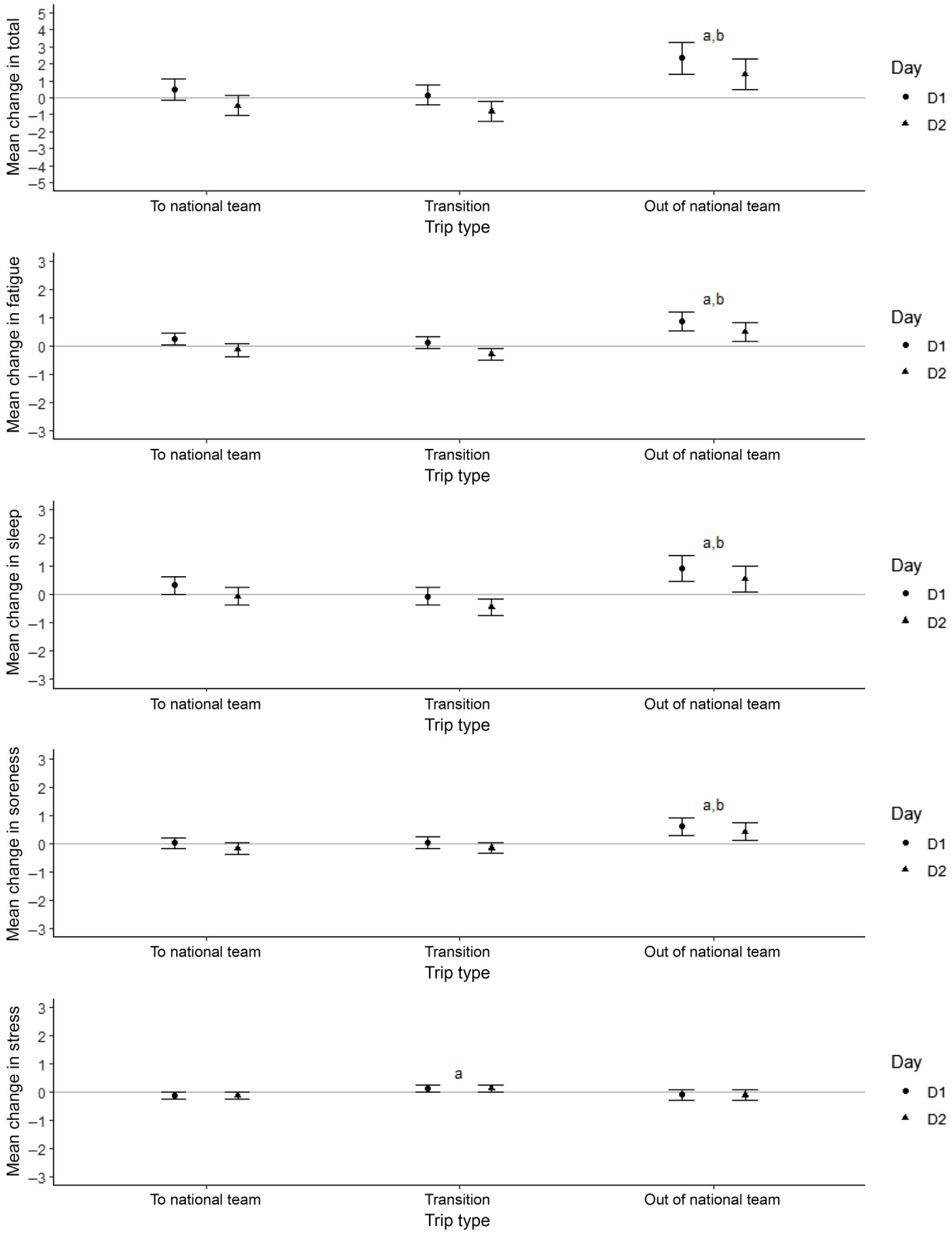

Figure 3 — Mean change in perceptual wellness scores by trip type (values are averaged out over levels of other variables). [a]Significantly different from travel to national team. [b]Significantly different from transition.

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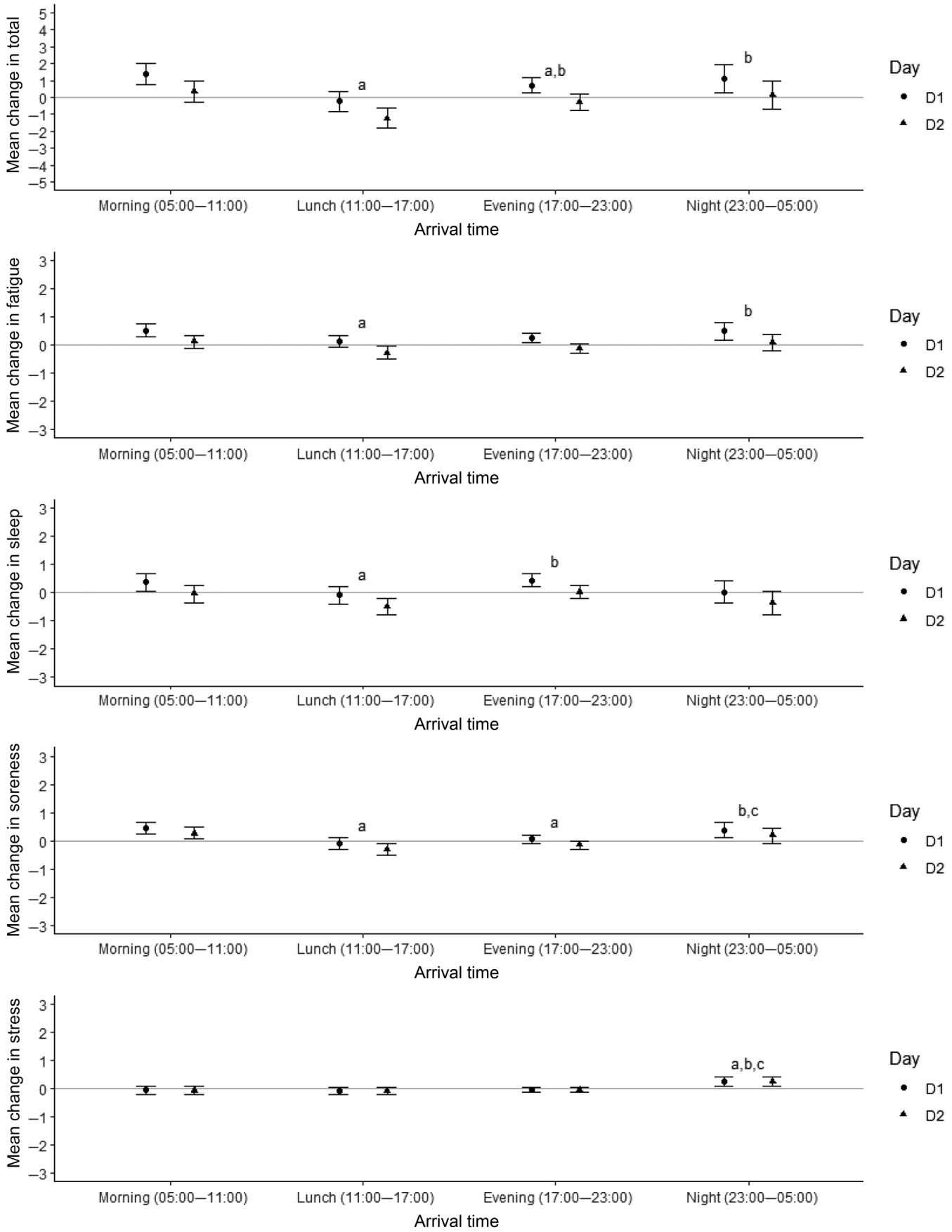

Figure 4 — Mean change in perceptual wellness scores by arrival time (values are averaged out over levels of other variables). [a]Significantly different from travel morning. [b]Significantly different from travel lunch. [c]Significantly different from travel evening.

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form of jet lag mitigation strategies[27] is recommended for athletes traveling from Europe to Australia or returning to Europe from Asia. Of note, elevated jet lag ratings were evident in older athletes and lower jet lag ratings in more experienced athletes. Prior studies have observed detrimental effects of age on jet lag symptoms,[10,28] however, such findings are not consistent, with studies reporting positive[18] or no effects of age.[11,29] The protective effect of experience has been previously observed among traveling footballers,[10,11] and development of travel management strategies is recommended for inexperienced players.

Although jet lag ratings likely provide a better indication of travel stress, perceptual wellness scales commonly collected in football teams showed some, albeit low, responsiveness to different travel bouts. Total wellness, fatigue, and sleep ratings were highest following travel from Europe to Australia. As fatigue and impaired sleep are common symptoms of jet lag/travel fatigue,[1] these elevated ratings are likely explained by the long-haul travel and eastward direction. Prior research assessing Europe to Oceania travel observed no changes in objective sleep measures in professional Rugby 7s athletes in the 6 days following arrival.[30] The contrasting findings may relate to differences in the sensitivity of objective versus subjective sleep measures; alongside, the authors suggesting travel management strategies implemented by the Rugby 7s athletes prevented sleep deficits.[30] All other flight paths included in this study appeared to have limited impact on perceptual responses, thus priority should be with players undertaking travel from Europe to Australia and on return to Europe from Asia.

Regardless of flight path, athletes reported higher perceptual fatigue, sleep, soreness, and total wellness scores after travel from the national team back to their club, which is a novel finding that has not been reported previously in national team athletes. Elevated jet lag and fatigue scores have been observed following the return journey of a round-trip domestic American travel schedule; however, comparisons were not reported between outbound and return travel.[17] As such, this study highlights that athletes may have additional difficulty in recovering from travel back to clubs following national team duties. The elevated ratings following return travel could be explained by effects of prior training/match load from the national team duties.[19] While data were not available for this study, it is likely that variations in physical load prior to travel will influence an athlete's well-being state[19] and therefore, may interact with posttravel perceptual responses. Further exploration is necessary to examine the interaction between prior match load and travel demands on athlete recovery. Also of concern are the short timeframes between national team and club matches and the frequent need to travel almost immediately following matches. Such requirements may restrict opportunities for rest and recovery interventions following matches and future research should explore how the time between match completion and travel departure influences postmatch travel responses. Accumulated travel fatigue from the short-term congested travel schedules (ie, national team athletes are often required to undertake up to 3 long-haul flights in 2 wk) may also partially explain the poorer responses to return travel.[26] However, given the lack of studies assessing responses of athletes to multiple long-haul trips in a short-time frame, this remains speculative. Regardless, this study highlights a need for travel and recovery interventions for athletes returning to their club following national duties.

An athlete's time of arrival should also be considered when traveling into and out of national teams. On the day after arrival, lower fatigue and sleep ratings were evident when arriving around daytime (11:00–17:00) compared with the morning or evening. These findings are similar to those of Waterhouse et al[18] who reported better jet lag and fatigue scores in athletes and support staff arriving late afternoon compared with early morning following travel from Europe to Australia. Those authors suggested the longer period of wakefulness for the morning arrival group may have induced greater fatigue ratings.[18] The findings of the current study may also be explained by the additional time for athletes to arrive at their hotel prior to attempting sleep and thus less interruption to the sleep period on the night of arrival. Where logistically possible, travel schedules arriving during the middle of the day to later afternoon are recommended; however, this may not be feasible and ensuring athletes are provided with adequate sleep on the night of arrival is important.

While the findings of this study provide useful insight into monitoring travel in national football teams, several limitations need to be considered. Although data collection for this study occurred in an ecologically valid national team environment, limited control existed across what the athletes did before, during, or after travel. Given the perceptual scales used are likely influenced by other external factors (evidenced by low R^2 values), caution should be taken on the application of study findings. The pretravel baseline used for comparisons represents the measure from a single day and, as such, may be more susceptible to external influences. Finally, perceptual jet lag ratings were obtained at a single time point during the day and variations in scores may be expected if the ratings were performed at other points throughout the day.[21]

Practical Applications

- In national-team footballers, subjective jet-lag ratings are more responsive to variations in travel demands than perceptual wellness scales.
- Europe to Australia or Asia to Europe travel resulted in reduced perceptual ratings for this national team, and thus additional travel-management strategies may be required.
- Additional support for national team footballers may be required when traveling back to clubs following national team camps or when arriving later in the day.

Conclusions

This study has identified several trip-related factors likely to cause poorer perceptual responses to travel in national-team footballers. Europe to Australia or Asia to Europe travel appears to be the most challenging to athletes from this national federation, and thus additional support may be required for these trips. Return travel after a national-team camp produced poorer ratings of fatigue, sleep, and soreness, and thus further support may be required when returning athletes to clubs. Daytime arrivals (11 AM to 5 PM) were the least detrimental to sleep and fatigue ratings and where possible trips should be scheduled to arrive during the day. Overall, the specific conditions of the trip should be considered, and travel-management strategies individualized when planning for the transport of players to and from national-team camps.

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